# Youssef Mohamed Helmy

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## Education

**Alexandria University** — Alexandria, Egypt
*Bachelor of Science in Computer Engineering — GPA: 3.7/4.0* — *Sep. 2022 – July 2027*

**Relevant Coursework**

- Data Structures
- Operating Systems
- Algorithms Analysis
- Computer Networks
- Database Management
- Computer Architecture
- Software Engineering
- OOP Design

## Projects

**URL Shortener API** | *Node.js, Express, MongoDB, Redis, JWT, Docker*

– Built a RESTful URL-shortening service with JWT access and refresh token authentication and per-user ownership control, restricting edit/delete/stats access to a link's original creator.
– Designed a Redis caching layer for URL lookups and stats alongside Redis-backed rate limiting on link creation, reducing repeated database reads under load.
– Wrote an automated test suite with Node's built-in test runner and Supertest, covering the full auth flow and URL lifecycle, using the GitHub Actions CI pipeline to run it against live MongoDB and Redis service containers.
– Containerized the service with Docker and Docker Compose for one-command local development against self-hosted MongoDB and Redis instances.

**Order Processing System** | *Node.js, Express, MySQL, RESTful APIs*

– Engineered a full-stack bookstore management system featuring separate customer and admin dashboards with RESTful API architecture.
– Implemented automated database triggers for real-time inventory restocking and negative stock prevention, ensuring data integrity across all transactions.
– Built an analytics and reporting module covering monthly/daily sales, top customers, and best-selling books to support data-driven business decisions.

**Connect Hub** | *Java, OOP, JSON Serialization, Custom Data Structures*

– Engineered a scalable social networking engine using core Java, focusing on robust backend logic for user relations, content distribution, and real-time state management.
– Developed a high-performance custom data access layer using JSON serialization to manage complex relational data without a traditional RDBMS.

## Technical Skills

**Languages:** Java , Python, JavaScript, C, C++, Dart

**Backend & Frameworks:** Spring Boot, Express.js, MongoDB, Redis, MySQL, JWT, RESTful APIs, JSON-based DB, Flutter

**Developer Tools:** Docker, Git, GitHub,GitHub Actions, Maven/Gradle, Android Studio, VS Code

**Core CS:** OOP, Multithreading, Design Patterns, SOLID Principles, System Architecture

## Extracurricular Activities

**Egyptian Collegiate Programming Contest (ECPC)** | *Contestant* — 2025, 2026

**Artificial Intelligence Course** | *WE* — Feb. 2024